SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING

Prasanthi Nilayam Campus

Department of Physics

INTERNSHIP REPORT

Synthesis and Structural Characterization of Lead-Free $K_{0.48}Na_{0.52}NbO_3$ (KNN) Piezoelectric Ceramics via Conventional Solid-State Reaction

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B.S. (Hons. with Research) Physics | Semester II

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Internship duration: 18 May 2026 – 18 June 2026

ACKNOWLEDGEMENTS

I express my sincere gratitude to Prof. Dibakar Das, Professor and Former Dean, School of Engineering Sciences & Technology, University of Hyderabad, for giving me the opportunity to train in his laboratory and for his guidance throughout the internship. I am thankful to the research scholars of the laboratory, who patiently taught me the practical side of solid-state synthesis, sintering and X-ray diffraction from first principles, and to the School of Engineering Sciences & Technology for the laboratory and X-ray diffraction facilities.

I thank the Department of Physics, Sri Sathya Sai Institute of Higher Learning, for permitting me to take up this internship, and my family and friends for their constant support.

With gratitude, I dedicate this work at the Lotus Feet of Bhagawan Sri Sathya Sai Baba.

ABSTRACT

This report documents a research internship on the synthesis of lead-free potassium sodium niobate ($K_{0.48}Na_{0.52}NbO_3$, KNN) piezoelectric ceramics by the conventional solid-state reaction route. Stoichiometric quantities of K_2CO_3, Na_2CO_3 and Nb_2O_5 (total precursor mass 5.6089 g) were wet-milled, split into two equal batches of about 2.785 g and calcined under different conditions: Batch 1 (B1) at 825 °C for 4 h and Batch 2 (B2) at 900 °C for 2 h. Green pellets were pressed in an 8 mm die at 2 tons (about 390 MPa), giving relative green densities (RGD) of 62–68%. Sintering experiments were carried out at 1000 °C, 1100 °C and 1150 °C.

Sintering B1 pellets at 1000 °C gave no appreciable densification (RSD about 63–64%). At 1100 °C for 4 h the relative sintered density rose to 88.73% for B1 and about 91.3–91.4% for B2. X-ray diffraction (XRD) of the calcined powders of both batches showed a single-phase orthorhombic perovskite structure, with B2 giving slightly sharper reflections; XRD of sintered pellets and the 1150 °C results are pending. The results show how strongly KNN densification depends on calcination conditions and sintering temperature, in line with published work on plain KNN.

Keywords: lead-free piezoelectrics; KNN; solid-state reaction; calcination; sintering; densification; X-ray diffraction

1. Introduction

Piezoelectric materials, which interconvert mechanical and electrical energy, underpin sensors, actuators, transducers, ultrasonic motors and energy-harvesting devices. For decades, lead zirconate titanate ($Pb(Zr,Ti)O_3$, PZT) has dominated the market because of its excellent electromechanical properties and mature fabrication technology. However, PZT contains approximately 60 wt.% lead, a hazardous heavy metal subject to increasing environmental regulation worldwide. This has driven intensive research into lead-free alternatives.

Among lead-free candidates, potassium sodium niobate ($(K,Na)NbO_3$, KNN)-based ceramics are among the most promising. KNN is a perovskite solid solution with a high Curie temperature (T_C about 400 °C), useful piezoelectric coefficients (d_{33} of 50–500 pC/N depending on composition and processing), and compatibility with conventional solid-state synthesis. Recent advances have come from phase boundary engineering (PBE), defect engineering and compositional modification, notably substitution with $Bi_{0.5}Na_{0.5}HfO_3$ (BNH), which creates an orthorhombic–tetragonal (O–T) phase boundary at room temperature and strongly enhances d_{33}.

The broader research direction of the laboratory is the development of KNN–BNH-based hard piezoceramics with simultaneously high d_{33} and mechanical quality factor Q, motivated by the strategy of embedding local Cu-acceptor defect dipoles in O–T phase boundary engineering reported by Qiu et al. (2025). As a prerequisite to this doped system, a thorough understanding of the synthesis and processing of the baseline plain KNN ceramic is essential.

The objective of this internship was therefore to synthesize and partially characterize plain $K_{0.48}Na_{0.52}NbO_3$ ceramics by the conventional solid-state reaction route, and to understand the effect of calcination conditions and sintering temperature on powder quality and pellet densification, which are the main processing variables controlling the final microstructure and properties.

2. Literature Review

2.1 Qiu et al. (2025), Nature Communications

The primary motivating reference is the 2025 Nature Communications paper by Qiu et al., "Excellent hardening effect in lead-free piezoceramics by embedding local Cu-doped defect dipoles in phase boundary engineering". It introduced a strategy for achieving high d_{33} and high Q simultaneously in KNN-based ceramics, two properties that normally trade off. The composition studied was $0.95(K_{0.48}Na_{0.52})NbO_3$–$0.05(Bi_{0.5}Na_{0.5})HfO_3$–0.2 mol% Fe_2O_3–x mol% CuO (KNN–BNH–xCu).

The mechanism involves two types of Cu^{2+}-related defect complexes. Dimeric complexes form defect dipole polarization that pins domain-wall motion and raises Q, while trimeric complexes introduce local structural heterogeneity, reduce domain size and flatten the Gibbs free-energy barrier at the O–T phase boundary, preserving high d_{33}. At the optimal doping (x = 1) the system reached d_{33} = 340 pC/N and Q = 256, a fourfold increase in Q for only a one-fifth reduction in

d_{33} relative to the undoped baseline. This paper shaped the understanding of defect dipoles, phase boundary engineering and sintering aids (Fe_2O_3, CuO) behind this internship.

2.2 Jiten et al. (2017), Materials Research Bulletin

Jiten et al. served as the primary synthesis-protocol reference. They investigated the effect of planetary ball-milling speed (100–400 rpm) on the microstructure and piezoelectric properties of plain $K_{0.5}Na_{0.5}NbO_3$, using a Fritsch Pulverisette 6 mill with zirconia media, ethanol as medium and an 8:1 ball-to-powder ratio. They found 325 rpm to be optimal, giving particle sizes of about 10 nm, a sintered density of about 95% and d_{33} of 114 pC/N after sintering at 1110 °C, with calcination at 900 °C for 4 h. Their milling speed, calcination temperature and sintering temperature were used as reference targets during this internship.

2.3 Verma et al. (2022), Journal of Materials Science: Materials in Electronics

Verma et al. studied the influence of calcination temperature (850–1050 °C) and sintering temperature (1075–1150 °C) on the structural, microstructural, dielectric, ferroelectric and piezoelectric properties of plain KNN prepared by solid-state reaction. Calcination at 1050 °C gave the most crystalline single-phase orthorhombic powder, shown by Rietveld refinement with the lowest goodness-of-fit (χ^2 = 2.61). Sintering at 1125 °C gave the best properties: relative density above 95%, average grain size of 3.33 μm, ε_r of about 460 at 10 kHz, remnant polarization P_r of about 98.21 μC/cm^2 and d_{33} of about 290 pC/N. Pellets sintered at 1100 °C or below had relative densities under 95% with visible porosity in SEM images, directly relevant to the densification challenges met in this work.

3. Experimental Work

3.1 Composition and stoichiometry

The target composition was $K_{0.48}Na_{0.52}NbO_3$, a slightly Na-rich perovskite that matches the KNN matrix of the Qiu et al. reference system. The solid-state reaction is:

$$0.24\ K_2CO_3 + 0.26\ Na_2CO_3 + 0.5\ Nb_2O_5 \rightarrow K_{0.48}Na_{0.52}NbO_3 + 0.5\ CO_2\uparrow$$

High-purity precursor powders were weighed for a total precursor mass of 5.6089 g, targeting about 5 g of KNN after CO_2 loss during calcination, on an analytical balance with ±0.0001 g accuracy (Table 1). The mixture was later divided into two equal portions of about 2.785 g each for separate calcination experiments.

Table 1: Precursor masses

Precursor	Mass (g)	Role
K_2CO_3	0.9679	K source
Na_2CO_3	0.8031	Na source
Nb_2O_5	3.8379	Nb source

3.2 Ball milling

The weighed powders were loaded into a planetary ball mill with ethanol as the milling medium, chosen over water to prevent dissolution of the highly water-soluble alkali carbonates. The ball-to-powder ratio (BPR) was 5:1. A single milling stage was performed on the combined mixture before splitting and calcination, to ensure homogeneous mixing and intimate particle contact. The slurry was dried and the dried cake lightly ground in an agate mortar and pestle before calcination.

3.3 Calcination

The milled and dried mixture was divided into two equal portions and calcined under different conditions to compare the effect of temperature and duration on powder quality (Table 2). Both batches were calcined in covered crucibles embedded in KNN pack powder to buffer alkali volatility. The temperatures were chosen from the literature range for KNN (typically 850–1050 °C), with B2 close to the 900 °C used by Jiten et al. After calcination, the powders were characterized by XRD (Section 4.3) and lightly ground to break up agglomerates before pressing.

Table 2: Calcination schedules

Batch	Temperature (°C)	Time (h)	Mass (g)
Batch 1 (B1)	825	4	≈ 2.785
Batch 2 (B2)	900	2	≈ 2.785

3.4 Pellet pressing

Calcined powders from each batch were mixed with 5 wt.% polyvinyl alcohol (PVA) binder solution and granulated. Green pellets were pressed uniaxially in an 8 mm stainless-steel die on a hydraulic press at 2 tons (about 390 MPa). The relative green density was calculated as RGD (%) = (ρ_{exp} / ρ_{theor}) × 100, where ρ_{exp} = M / V, V = (π/4) D^2 T, and the theoretical density of KNN was taken as 4.48 g/cm^3. Six pellets were pressed from B1 and five from B2 (Tables 3 and 4).

Table 3: Green pellet data, B1 (calcined at 825 °C / 4 h)

Pellet	Mass (g)	Dia (mm)	Thickness (mm)	RGD (%)
1	0.2834	8.04	1.974	64.0
2	0.2747	8.04	1.928	63.5
3	0.2765	8.03	1.999	62.4
4	0.2786	8.04	1.928	64.4
5	0.2810	8.03	1.932	65.6
6	0.2799	8.04	1.950	64.1

Table 4: Green pellet data, B2 (calcined at 900 °C / 2 h)

Pellet	Mass (g)	Dia (mm)	Thickness (mm)	RGD (%)
1	0.2818	8.08	1.79	68.2
2	0.2798	8.08	1.92	63.1
3	0.2808	8.08	1.88	64.7
4	0.2795	8.10	1.81	66.87
5	0.2804	8.07	1.84	66.54

3.5 Sintering

Before sintering, the green pellets underwent binder burnout at 600 °C for 4 h to remove the PVA. The sintering profile had an intermediate hold at 850 °C followed by a ramp to the target temperature. All pellets were sintered in covered alumina crucibles embedded in KNN pack powder to suppress alkali (K, Na) volatilization.

Sintering was done at three conditions across the two batches: 1000 °C/2 h (B1 pellets 1–3), 1100 °C/4 h (B1 pellet 6; B2 pellets 2–3) and 1150 °C/4 h (B2 pellets 4–5, results pending). The 1000 °C condition was the initial attempt, chosen with caution about alkali volatility; it proved insufficient for meaningful densification of plain KNN, consistent with Verma et al. (2022).

3.6 Density measurement

After sintering, the relative sintered density was determined geometrically from the measured mass and dimensions (diameter and thickness with digital calipers): RSD (%) = ($\rho_{sintered}$ / ρ_{theor}) × 100. The data for all measured pellets are given in Tables 5 and 6.

Table 5: Sintered pellet data, B1

Pellet	Sintering condition	Dia (mm)	Thickness (mm)	Mass (g)	RSD (%)
1	1000 °C / 2 h	7.98	1.966	0.2780	63.2
2	1000 °C / 2 h	7.94	1.925	0.2654	64.2
3	1000 °C / 2 h	7.95	1.991	0.2740	63.8
6	1100 °C / 4 h	7.08	1.770	0.2769	88.73

Table 6: Sintered pellet data, B2

Pellet	Sintering condition	Dia (mm)	Thickness (mm)	Mass (g)	RSD (%)
2	1100 °C / 4 h	7.23	1.640	0.2758	91.40
3	1100 °C / 4 h	7.21	1.689	0.2747	91.27
4	1150 °C / 4 h	pending	pending	pending	pending
5	1150 °C / 4 h	pending	pending	pending	pending

4. Results and Discussion

4.1 Green density

All green pellets were pressed at 2 tons (about 390 MPa) in the 8 mm die. B1 pellets reached RGD between 62.4% and 65.6%, while B2 pellets were slightly higher (63.1–68.2%), with pellet 1 at 68.2% the highest of all (Figure 1). The difference between batches is likely due to differences in powder compressibility from the different calcination conditions; the 900 °C powder may have had a different particle morphology and agglomerate hardness from the 825 °C powder.

Very high green density (above about 68%) risks trapping residual gases during sintering and creating closed porosity that hinders final densification. The observed values are generally acceptable for KNN prepared by conventional pressing, and the consistency within each batch shows reproducible pressing.

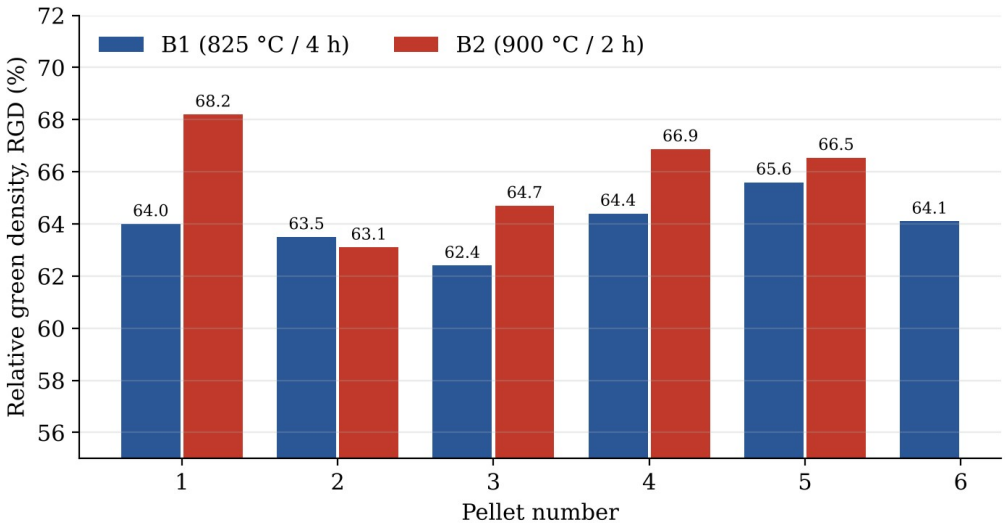


Figure 1: Relative green density of all pressed pellets, B1 and B2.

4.2 Sintering and densification

The main finding is the strong dependence of densification on sintering temperature (Figures 2 and 3). B1 pellets sintered at 1000 °C/2 h showed essentially no densification: their RSD of 63.2%, 64.2% and 63.8% is similar to the green values of about 62–65%. This shows 1000 °C is well below the effective sintering threshold for plain KNN, which typically needs temperatures above 1100 °C for grain bonding and pore elimination, consistent with Verma et al. (2022), who saw visible porosity and relative density below 95% at lower sintering temperatures.

At 1100 °C/4 h the improvement was marked. B1 pellet 6 reached 88.73%, and B2 pellets 2 and 3 reached 91.40% and 91.27%, higher than B1 at the same condition. This likely reflects better powder quality for B2 (calcined at 900 °C instead of 825 °C), giving a more completely reacted, phase-pure and more sinterable KNN powder.

Although about 91% RSD for B2 is a large improvement over the failed 1000 °C batch, it is still below the roughly 95% generally needed for reliable functional characterization of

piezoelectric ceramics. This agrees with reports that plain KNN, without liquid-phase sintering aids, needs 1100–1150 °C or higher for full densification. The B2 pellets sintered at 1150 °C/4 h (pending) should show higher RSD and give a direct comparison of the effect of sintering temperature within one batch.

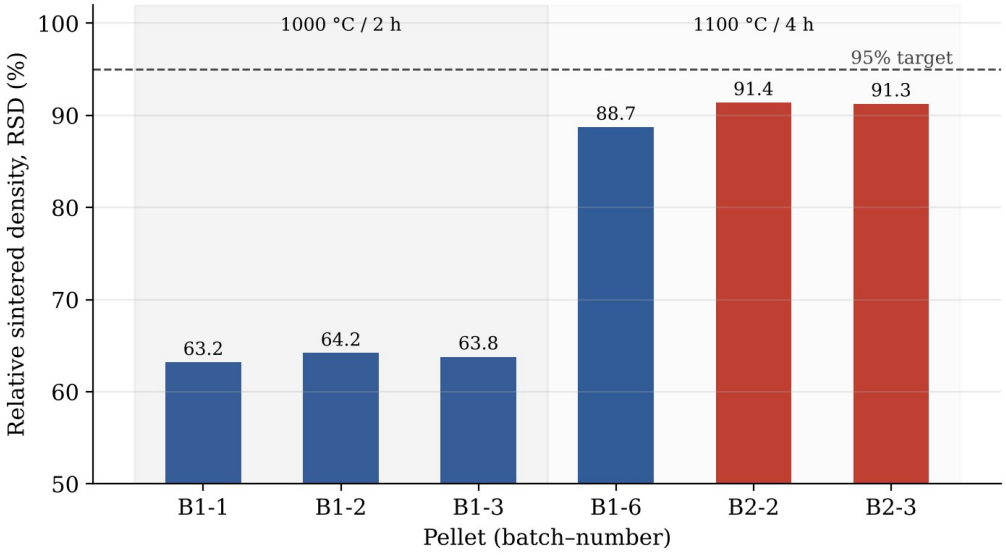


Figure 2: Relative sintered density of the measured pellets against the 95% target.

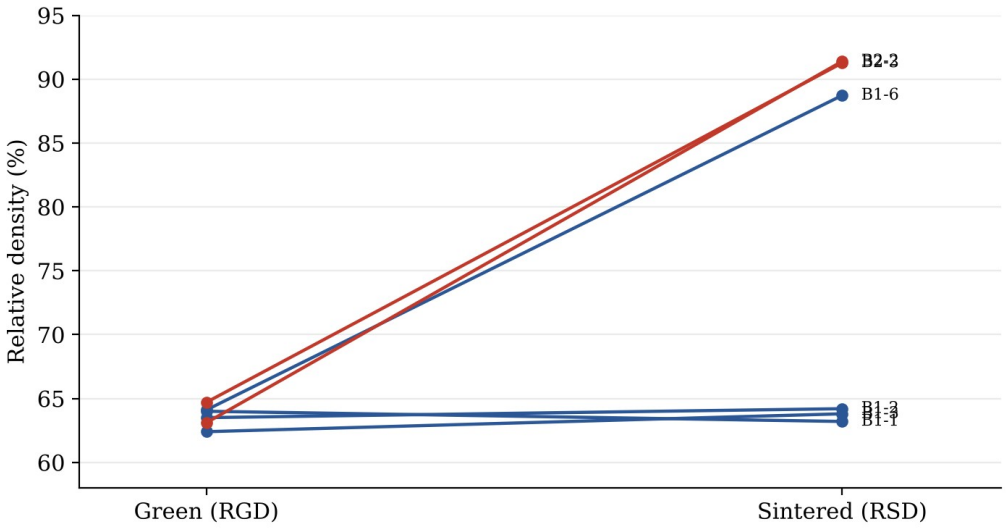


Figure 3: Change in relative density from green (RGD) to sintered (RSD) for each measured pellet.

4.3 XRD characterization

XRD patterns of the calcined powders of B1 (825 °C/4 h) and B2 (900 °C/2 h) were collected on a Bruker D8 Advance diffractometer (Cu Kα radiation) over 2θ = 20–60°. The raw scans were background-subtracted and $K\alpha_2$-stripped before analysis. The patterns are shown in Figure 4.

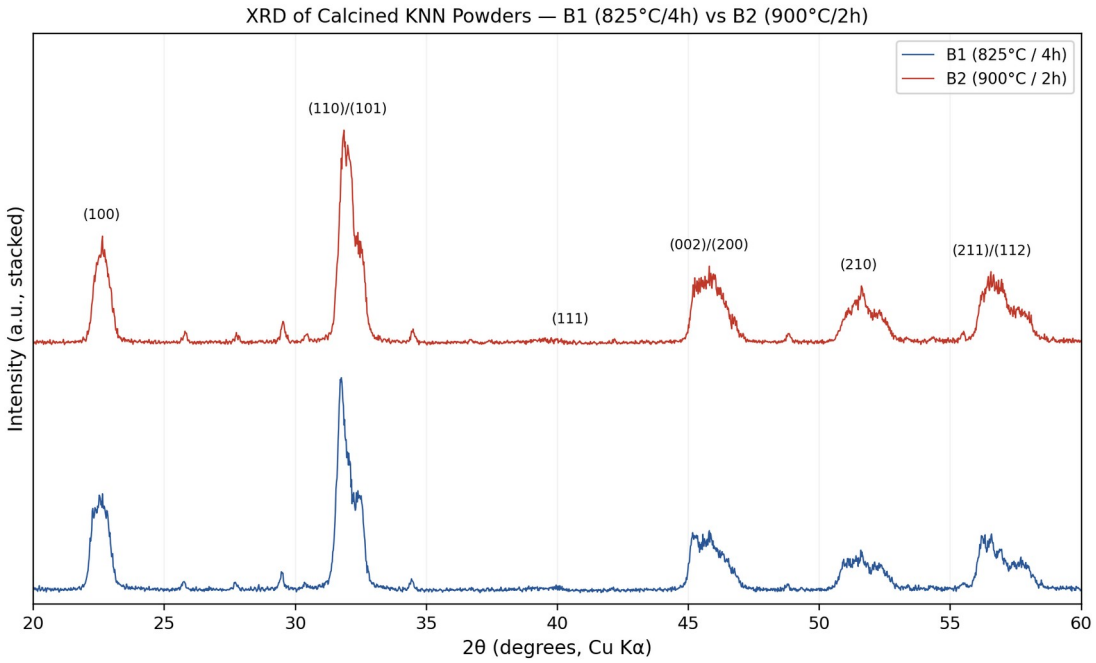


Figure 4: XRD patterns of calcined KNN powders, B1 (825 °C/4 h, bottom) and B2 (900 °C/2 h, top), stacked with a vertical offset. Miller indices assigned by comparison with the orthorhombic KNN reference pattern (JCPDS No. 98-024-7572).

Both patterns show the characteristic single-phase perovskite reflection sequence of KNN: the (100) peak near 2θ ≈ 22.5–22.7°, the dominant (110)/(101) reflection near 31.8–32.4°, and higher-angle reflections near 45–46°, 51–52° and 56–57°. No strong peaks from residual K_2CO_3, Na_2CO_3 or Nb_2O_5 are seen in either pattern, indicating that the solid-state reaction went essentially to completion under both calcination conditions. A few very weak reflections near 2θ ≈ 25.8°, 27.7°, 29.5° and 34.5° appear in both patterns; they are not assigned here and would be examined during Rietveld refinement as possible traces of a minor secondary phase.

The diagnostic {200}/{002} pseudocubic doublet, used to distinguish orthorhombic from tetragonal KNN, is resolved as two overlapping maxima between 2θ ≈ 45° and 46° in both patterns. This confirms an orthorhombic perovskite structure (space group Amm2) at room temperature, consistent with Jiten et al. (2017) and Verma et al. (2022).

A qualitative comparison shows that the B2 (900 °C/2 h) pattern has a somewhat sharper and more intense (110)/(101) reflection than B1, suggesting marginally better crystallinity at the higher calcination temperature. This is consistent with the higher RSD of B2 pellets (91.3–91.4%) than the comparable B1 pellet (88.73%) at the same sintering condition. No Rietveld refinement, Scherrer crystallite-size or lattice-parameter analysis has yet been done on this data; these are immediate next steps (Section 6).

XRD of the sintered pellets (B1 pellet 6 and B2 pellets 2–3) is pending. It will confirm phase purity after sintering and allow comparison of peak sharpness and any peak shifts that would indicate lattice-parameter changes from alkali volatility at high temperature.

5. Skills Developed

5.1 Laboratory techniques

- Stoichiometric calculation for multi-component ceramic systems from a target batch mass, and accurate weighing of sub-gram quantities on an analytical balance.
- Planetary ball milling (BPR 5:1, wet milling in ethanol, slurry recovery and drying) and agate-mortar grinding to break agglomerates.
- Calcination protocol: crucible preparation, pack-powder technique and furnace programming.
- Uniaxial pellet pressing, caliper measurement and calculation of RGD and RSD.
- Multi-stage sintering: binder burnout and high-temperature sintering profiles.
- XRD sample preparation, data reduction (background subtraction, $K\alpha_2$ stripping) and pattern analysis.

5.2 Conceptual understanding

- Perovskite crystal structure and the crystallographic basis of piezoelectricity in KNN; orthorhombic versus tetragonal KNN from the {200} pseudocubic doublet.
- Solid-state reaction mechanism: precursor decomposition, diffusion-driven phase formation and CO_2 evolution.
- Role of calcination temperature and time in phase purity, and the relation between green density and sintered density.
- Sintering mechanisms (grain-boundary diffusion, pore elimination, grain growth) and alkali volatility in KNN with its mitigation (pack powder, covered crucibles).
- Defect-dipole mechanism in Cu-doped KNN (dimeric versus trimeric complexes) and the d_{33}–Q trade-off; principle of Rietveld refinement (MAUD, GSAS-II).

5.3 Research and communication

- Reading primary materials-science literature critically and replicating synthesis protocols from published papers.
- Systematic laboratory record-keeping, data tabulation and trend interpretation, and scientific report writing.

6. Conclusion and Future Work

This internship gave hands-on experience of the complete solid-state synthesis pipeline for lead-free KNN piezoelectric ceramics, from stoichiometric calculation through calcination, pellet pressing and sintering. Two batches of plain $K_{0.48}Na_{0.52}NbO_3$ were synthesized under different calcination conditions (825 °C/4 h and 900 °C/2 h), and the effect of sintering temperature on densification was explored.

The key finding is that sintering temperature is the dominant variable controlling densification of plain KNN: 1000 °C is insufficient (RSD about 63–64%), while 1100 °C/4 h gives much better densification (RSD about 89–91%), with the higher-temperature-calcined B2 batch

outperforming B1. XRD of the calcined powders shows both batches to be single-phase orthorhombic KNN, with B2 showing marginally sharper reflections consistent with its better sinterability. Sintering at 1150 °C/4 h is in progress and is expected to push the density closer to the 95% needed for functional characterization.

The next steps are: (1) collection and analysis of sintered-pellet XRD data; (2) Rietveld refinement and Scherrer crystallite-size analysis of the calcined-powder XRD data; (3) surface polishing and silver electrode deposition on the best-density pellets; (4) poling in silicone oil at about 3 kV/mm; and (5) measurement of d_{33}, k_p and Q as a baseline for the KNN system. In the longer term, the doped KNN–BNH–CuO/Fe_2O_3 composition of Qiu et al. (2025), with its engineered O–T phase boundary and defect-dipole hardening, is the primary target of the laboratory.

The internship gave a firm foundation in ceramic processing science and direct exposure to an active area of functional-materials research.

7. References

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